

Statistics: Mean & Median — Full Solutions

Digital SAT Style | Every Step Fully Explained

Q1. Correct Answer: B) 20

STEP-BY-STEP SOLUTION:

Sorted: 12, 15, 18, 20, 22, 22, 31.

7 values → median = 4th value = 20.

WHY EACH OPTION IS WRONG/RIGHT:

A) 18 — 3rd value, not 4th.

B) 20 — CORRECT.

C) 22 — 5th value.

D) 24 — Not a data value.



TRAP ALERT: Counting from the wrong end or confusing median with mean.



KEY INSIGHT: For n values: median position = $(n+1)/2$. For 7: position 4.



FAST METHOD: 7 vals. Middle=4th=20.

Q2. Correct Answer: B) 10

STEP-BY-STEP SOLUTION:

Sum = $4+9+9+13+15 = 50$

Mean = $50/5 = 10$

WHY EACH OPTION IS WRONG/RIGHT:

A) 9 — This is the median (middle value), not the mean.

B) 10 — CORRECT.

C) 11 — Arithmetic error.

D) 13 — Just the 4th value.



TRAP ALERT: Confusing mean with median. Median=9 (middle), Mean=10 (average). Different!



KEY INSIGHT: Mean = sum ÷ count. Median = middle value when sorted.



FAST METHOD: $50/5=10$.

Q3. Correct Answer: A) Mean > Median

STEP-BY-STEP SOLUTION:

Sorted: 6, 7, 7, 8, 12. Median=7 (3rd of 5).

Mean = $(6+7+7+8+12)/5 = 40/5 = 8$.

$8 > 7$ → Mean > Median.

WHY EACH OPTION IS WRONG/RIGHT:

A) CORRECT.

B) $8 \neq 7$, so not equal.

C) Mean is larger, not smaller.

D) Median=7 (3rd of 5 values, no averaging).

 **TRAP ALERT:** The outlier 12 pulls the mean up. Mean > Median when data is right-skewed.

 **KEY INSIGHT:** Right-skewed data (outlier on high end): Mean > Median. Left-skewed: Mean < Median.

 **FAST METHOD:** Mean=8, Median=7. $8 > 7 \rightarrow A$.

Q4. Correct Answer: C) 65

STEP-BY-STEP SOLUTION:

6 values (even count) $\rightarrow$ median = average of 3rd and 4th.

3rd=64, 4th=66. Median=(64+66)/2=65.

WHY EACH OPTION IS WRONG/RIGHT:

A) 63 — Average of wrong pair.

B) 64 — 3rd value only.

C) 65 — CORRECT.

D) 66 — 4th value only.

 **TRAP ALERT:** Even-count data: you MUST average the two middle values. Taking just one is wrong.

 **KEY INSIGHT:** Even n: median = average of $(n/2)$ th and $(n/2+1)$ th values.

 **FAST METHOD:** 6 vals $\rightarrow$ avg of 3rd(64) and 4th(66)=65.

Q5. Correct Answer: 84

STEP-BY-STEP SOLUTION:

Sorted: 72,80,84,88,96.

5 values $\rightarrow$ median = 3rd = 84.

WHY EACH OPTION IS WRONG/RIGHT:

80 — 2nd value.

85 — Averaged 84+86 (non-existent).

88 — 4th value.

84 — CORRECT.

 **TRAP ALERT:** With 5 values, some students average 2nd and 3rd as if there were 4.

 **KEY INSIGHT:** Odd count $\rightarrow$ single middle value. No averaging.

 **FAST METHOD:** 5 vals. Middle=3rd=84.

Q6. Correct Answer: B) 12

STEP-BY-STEP SOLUTION:

Mean=10, $n=5 \rightarrow$ sum=50.

Known: $6+8+10+14=38$.

$x = 50 - 38 = 12$.

WHY EACH OPTION IS WRONG/RIGHT:

- A) 10 — Sum=48, mean=9.6. Wrong.
- B) 12 — CORRECT.
- C) 14 — Sum=52, mean=10.4.
- D) 16 — Sum=54, mean=10.8.

 **TRAP ALERT:** Guessing $x=10$ because the mean is 10.

 **KEY INSIGHT:** Missing value = (target mean $\times$ n) – sum of known values.

 **FAST METHOD:** $50-38=12$.

Q7. Correct Answer: B) Median A < Median B

STEP-BY-STEP SOLUTION:

A: 5 vals, median=3rd=7.

B: 6 vals, median=avg(3rd,4th)=(7+9)/2=8.

$7 < 8 \rightarrow$ Median A < Median B.

WHY EACH OPTION IS WRONG/RIGHT:

A) $7 > 8$ is false.

B) CORRECT.

C) $7 \neq 8$.

D) We can calculate both precisely.

 **TRAP ALERT:** Assuming adding a high value (25) doesn't change the median. It shifts which values are "middle" when n changes.

 **KEY INSIGHT:** Adding a value above the current median can shift the median up when the count changes parity.

 **FAST METHOD:** A: median=7. B: median=(7+9)/2=8. $A < B$.

Q8. Correct Answer: C) 16

STEP-BY-STEP SOLUTION:

Mean=15, $n=4 \rightarrow$ sum=60.

Known: $12+14+18=44$.

4th = $60-44 = 16$.

WHY EACH OPTION IS WRONG/RIGHT:

A) 14 — Sum=58, mean=14.5.

B) 15 — Sum=59, mean=14.75.

C) 16 — CORRECT.

D) 17 — Sum=61, mean=15.25.

 **TRAP ALERT:** Assuming the missing value equals the mean (15). Only true in specific cases.

 **KEY INSIGHT:** Missing value = (mean $\times$ n) – known sum.

 **FAST METHOD:** $60-44=16$.

Q9. Correct Answer: B) 7

STEP-BY-STEP SOLUTION:

Sorted: 5,5,6,8,11,13.

6 values → avg of 3rd(6) and 4th(8).

$$(6+8)/2 = 7.$$

WHY EACH OPTION IS WRONG/RIGHT:

A) 6 — Only 3rd value.

B) 7 — CORRECT.

C) 8 — Only 4th value.

D) 9 — Arithmetic error.



TRAP ALERT: Taking one middle value instead of averaging both for even-count data.



KEY INSIGHT: Even count: ALWAYS average the two middle values.



FAST METHOD: $(6+8)/2=7$.

Q10. Correct Answer: C) 3rd

STEP-BY-STEP SOLUTION:

5 values: positions 1,2,3,4,5.

$$\text{Median} = (5+1)/2 = 3\text{rd position.}$$

WHY EACH OPTION IS WRONG/RIGHT:

A) 1st — The minimum.

B) 2nd — One below median.

C) 3rd — CORRECT.

D) 4th — One above median.



TRAP ALERT: Using $(n/2)$ instead of $(n+1)/2$ for odd n .



KEY INSIGHT: Median position for n values: $(n+1)/2$.



FAST METHOD: $(5+1)/2=3$.

Q11. Correct Answer: B) 6

STEP-BY-STEP SOLUTION:

Old: 8 numbers, mean=24 → sum=192.

New: 9 numbers, mean=22 → sum=198.

$$9\text{th number} = 198 - 192 = 6.$$

WHY EACH OPTION IS WRONG/RIGHT:

A) 4 → sum=196, mean≈21.8.

B) 6 — CORRECT.

C) 8 → sum=200, mean≈22.2.

D) 10 → sum=202, mean≈22.4.



TRAP ALERT: Subtracting means: $24-22=2$, then using 2 as the answer.



KEY INSIGHT: New value = new total – old total. Total = mean × count.



FAST METHOD: $198-192=6$.

Q12. Correct Answer: C) Both I and II

STEP-BY-STEP SOLUTION:

Original (6 vals): $\text{mean} = 63/6 = 10.5$, $\text{median} = (9+12)/2 = 10.5$.

Remove 18 (5 vals): $\text{mean} = 45/5 = 9$ ($\downarrow$). $\text{Median} = 3\text{rd} = 9$ ($\downarrow$).

Both decrease.

WHY EACH OPTION IS WRONG/RIGHT:

A) I only — Median also decreases.

B) II only — Mean also decreases.

C) CORRECT.

D) Both change.

 **TRAP ALERT:** Assuming removing the largest value doesn't affect the median. Changing n from 6(even) to 5(odd) redefines which position is the median.

 **KEY INSIGHT:** Removing any value changes $n \rightarrow$ may redefine the median position. Always recount.

 **FAST METHOD:** Old median = 10.5. New median = 9. Both drop $\rightarrow$ C.

Q13. Correct Answer: 28

STEP-BY-STEP SOLUTION:

6 values, median = average of 3rd and 4th.

$$(a+b)/2 = 14 \rightarrow a+b = 28.$$

WHY EACH OPTION IS WRONG/RIGHT:

14 — Wrote $a=b=14$ but answered 14 instead of sum.

22 — Arithmetic error.

25 — Different error.

28 — CORRECT.

 **TRAP ALERT:** The question asks for $a+b$, not for a or b individually.

 **KEY INSIGHT:** Even-set median = average of two middle values. Sum of those two = $2 \times$ median.

 **FAST METHOD:** $a+b = 2 \times 14 = 28$.

Q14. Correct Answer: B) 2.8

STEP-BY-STEP SOLUTION:

Weighted sum: $1(4)+2(6)+3(8)+4(5)+5(2)$

$$= 4+12+24+20+10 = 70$$

Total students: $4+6+8+5+2 = 25$

$$\text{Mean} = 70/25 = 2.8$$

WHY EACH OPTION IS WRONG/RIGHT:

A) 2.4 — Used unweighted approach.

B) 2.8 — CORRECT.

C) 3.0 — This is the mode (3 appears most).

D) 3.2 — Overestimate.

 **TRAP ALERT:** Averaging 1,2,3,4,5 unweighted ($=3.0$). You MUST weight by frequency.

💡 **KEY INSIGHT:** Frequency table mean: $\Sigma(\text{value} \times \text{frequency}) \div \Sigma(\text{frequency})$.

⚡ **FAST METHOD:** $70/25=2.8$.

Q15. Correct Answer: C) Median equals 28

STEP-BY-STEP SOLUTION:

Old: 7 vals, mean=30 $\rightarrow$ sum=210, median=28.

Add 28: new sum=238, n=8, mean=238/8=29.75<30.

Old median was 28 (4th of 7). Adding 28 near the middle:

New set (8 vals) $\rightarrow$ median=avg(4th,5th). Since 28 was already at position 4, adding another 28 keeps both middle positions at or near 28.

New median = 28.

WHY EACH OPTION IS WRONG/RIGHT:

A) Mean=29.75 $\neq$ 30.

B) True (29.75<30), but C is also must-be-true and more specific.

C) CORRECT: adding the median value preserves the median.

D) New median stays 28, not greater.

🏠 **TRAP ALERT:** Thinking adding 28 raises the median. Adding a value EQUAL to the current median preserves it.

💡 **KEY INSIGHT:** Adding a value equal to the current median $\rightarrow$ new median stays the same.

⚡ **FAST METHOD:** New mean<30 (since 28<30). Median stays 28. Answer C.

Q16. Correct Answer: 18

STEP-BY-STEP SOLUTION:

Sort: 8, 9, 12, 14, 22, 22, 31, 35.

8 values $\rightarrow$ avg of 4th and 5th.

$(14+22)/2 = 36/2 = 18$.

WHY EACH OPTION IS WRONG/RIGHT:

22 — Only 5th value.

14 — Only 4th value.

20 — Estimated without sorting.

18 — CORRECT.

🏠 **TRAP ALERT:** Not sorting the data first! Unsorted, positions 4 and 5 give wrong values.

💡 **KEY INSIGHT:** ALWAYS sort data before finding the median. #1 statistics error.

⚡ **FAST METHOD:** Sort $\rightarrow$ 8 vals $\rightarrow$ avg(14,22)=18.

Q17. Correct Answer: A) Both decrease

STEP-BY-STEP SOLUTION:

Original (10 vals): mean=64/10=6.4. Median=(5+7)/2=6.

Remove 12 (9 vals): mean=52/9 $\approx$ 5.78 ($\downarrow$). Median=5th=5 ($\downarrow$).

Both decrease.

WHY EACH OPTION IS WRONG/RIGHT:

- A) CORRECT.
- B) Median also changes from 6 to 5.
- C) Mean also changes.
- D) Both change.

 **TRAP ALERT:** Assuming removing the maximum only affects the mean. Changing n from 10(even) to 9(odd) redefines the median.

 **KEY INSIGHT:** Removing a value always changes n . Recalculate median position for the new n .

 **FAST METHOD:** Old median=6, new median=5. Both drop → A.

Q18. Correct Answer: C) 15

STEP-BY-STEP SOLUTION:

If $k \geq 12$: sorted as 6,9,12,k,18. Median=12 (3rd of 5).

Set mean=median: $(45+k)/5=12 \rightarrow 45+k=60 \rightarrow k=15$.

Verify sort: 6,9,12,15,18 ✓. Mean= $60/5=12$ =median ✓.

WHY EACH OPTION IS WRONG/RIGHT:

- A) 10 → sorted: 6,9,10,12,18. Median=10, mean= $55/5=11$. Not equal.
- B) 12 → sorted: 6,9,12,12,18. Median=12, mean= $57/5=11.4$. Not equal.
- C) 15 — CORRECT.
- D) 18 → sorted: 6,9,12,18,18. Median=12, mean= $63/5=12.6$. Not equal.

 **TRAP ALERT:** Assuming median is always 12 regardless of k . If $k < 12$, the sort order changes.

 **KEY INSIGHT:** When data contains an unknown, determine its sorted position first, then set up equations.

 **FAST METHOD:** Assume $k \geq 12 \rightarrow$ median=12. $(45+k)/5=12 \rightarrow k=15$. Check $15 \geq 12$ ✓.

Q19. Correct Answer: A) I only

STEP-BY-STEP SOLUTION:

Median=35 (3rd of 5). Mean= $200/5=40$.

I: $40 > 35 \rightarrow$ TRUE.

Remove 70 → {20,35,35,40}. New median= $(35+35)/2=35$. Same!

II: FALSE (median unchanged).

WHY EACH OPTION IS WRONG/RIGHT:

- A) CORRECT.
- B) II is false.
- C) Only I is true.
- D) I is true.

 **TRAP ALERT:** Assuming removing the max always lowers the median. Here both middle values are 35, so median stays 35.

 **KEY INSIGHT:** Outliers affect the MEAN, not the MEDIAN. This is a core DSAT concept.

 **FAST METHOD:** Mean= $40 >$ Median=35 → I true. Remove 70 → median still 35 → II false. A.

Q20. Correct Answer: 20**STEP-BY-STEP SOLUTION:**

7 consecutive even integers with mean=20.

Symmetric sequence: mean=median=middle value.

Middle (4th) value = 20.

Sequence: 14,16,18,20,22,24,26. Mean=140/7=20 ✓.

WHY EACH OPTION IS WRONG/RIGHT:

22 — Wrong position.

14 — First value.

26 — Last value.

20 — CORRECT.



TRAP ALERT: Overcomplicating it. For symmetric data, mean = median = center value.



KEY INSIGHT: Equally spaced data (arithmetic sequence): mean = median = middle value.



FAST METHOD: Symmetric → mean=median=20.

Q21. Correct Answer: C) Mean=18, Median=18**STEP-BY-STEP SOLUTION:**

Add 4 to each: P becomes {14,16,18,20,22}.

New mean = $14+4 = 18$.

New median = $14+4 = 18$.

WHY EACH OPTION IS WRONG/RIGHT:

A) No shift — wrong.

B) Only mean shifts — wrong, both shift.

C) CORRECT.

D) Only median shifts — wrong.



TRAP ALERT: Thinking the median doesn't shift. Adding a constant to EVERY value shifts both mean and median by that constant.



KEY INSIGHT: Adding constant c : new mean = old mean + c . New median = old median + c . Range and standard deviation stay the same.



FAST METHOD: Both shift by 4: $14+4=18$.

Q22. Correct Answer: A) \$4.00**STEP-BY-STEP SOLUTION:**

Sum = $4.20 \times 5 = \$21.00$.

Known: $3.00+4.00+4.50+5.50 = \$17.00$.

5th = $21.00-17.00 = \$4.00$.

WHY EACH OPTION IS WRONG/RIGHT:

A) \$4.00 — CORRECT.

B) \$4.20 — Copied the mean as the answer.

C) \$4.50 — Already in the set.

D) \$5.00 — Arithmetic error.

 **TRAP ALERT:** Picking \$4.20 because the mean is \$4.20. The missing value doesn't have to equal the mean.

 **KEY INSIGHT:** Missing value = $(\text{mean} \times n) - \text{known sum}$.

 **FAST METHOD:** $21 - 17 = \$4.00$.

Q23. Correct Answer: D) 12

STEP-BY-STEP SOLUTION:

Original: n values, $\text{mean} = 40 \rightarrow \text{sum} = 40n$.

Add 3 values of 10: $\text{new sum} = 40n + 30$, $\text{new count} = n + 3$.

New mean = 34: $(40n + 30)/(n + 3) = 34$

$40n + 30 = 34n + 102$

$6n = 72 \rightarrow n = 12$.

Verify: $\text{sum} = 480$. New $\text{sum} = 510$, new $n = 15$. $510/15 = 34$ ✓.

WHY EACH OPTION IS WRONG/RIGHT:

A) 3 $\rightarrow \text{sum} = 120$, $\text{new} = 150/6 = 25 \neq 34$.

B) 6 $\rightarrow \text{sum} = 240$, $\text{new} = 270/9 = 30 \neq 34$.

C) 9 $\rightarrow \text{sum} = 360$, $\text{new} = 390/12 = 32.5 \neq 34$.

D) 12 — CORRECT.

 **TRAP ALERT:** Setting up the equation wrong: forgetting that both numerator AND denominator change.

 **KEY INSIGHT:** New mean = $(\text{old sum} + \text{added values}) / (\text{old count} + \text{added count})$. Set up algebraically.

 **FAST METHOD:** $(40n + 30)/(n + 3) = 34 \rightarrow 6n = 72 \rightarrow n = 12$.

Q24. Correct Answer: 14

STEP-BY-STEP SOLUTION:

Data: 5, 8, 11, x , 17, 20 (sorted, $11 \leq x \leq 17$).

6 vals $\rightarrow \text{median} = (11 + x)/2$.

Mean = $(5 + 8 + 11 + x + 17 + 20)/6 = (61 + x)/6$.

Set equal: $(11 + x)/2 = (61 + x)/6$

Cross multiply: $3(11 + x) = 61 + x$

$33 + 3x = 61 + x \rightarrow 2x = 28 \rightarrow x = 14$.

Verify: $\text{median} = (11 + 14)/2 = 12.5$. Mean = $75/6 = 12.5$ ✓. $11 \leq 14 \leq 17$ ✓.

WHY EACH OPTION IS WRONG/RIGHT:

12 — Set median = 12 without using x .

16 — Arithmetic error.

11 — Used boundary.

14 — CORRECT.

 **TRAP ALERT:** Not including x in BOTH the mean and median expressions.

 **KEY INSIGHT:** Unknown in data set $\rightarrow$ appears in both mean and median formulas. Set up both with x and equate.

⚡ **FAST METHOD:** $3(11+x)=61+x \rightarrow 2x=28 \rightarrow x=14$.

Q25. Correct Answer: B)

STEP-BY-STEP SOLUTION:

Remove outlier 120 from 10-value set (mean=50, median=48).

New sum= $500-120=380$. New mean= $380/9 \approx 42.2$. Significant decrease.

Median of 10 vals=avg(5th,6th). 120 is the 10th. Removing shifts to 9 vals, median=5th.

The 5th value is essentially unchanged.

Mean drops significantly; median barely changes.

WHY EACH OPTION IS WRONG/RIGHT:

A) Median is robust — doesn't decrease significantly.

B) CORRECT.

C) Mean always changes when a value is removed.

D) Mean always changes.

👤 **TRAP ALERT:** Thinking removing any large value also significantly changes the median.

💡 **KEY INSIGHT:** Mean = sensitive to outliers (non-robust). Median = resistant to outliers (robust). Core DSAT concept.

⚡ **FAST METHOD:** Remove outlier $\rightarrow$ mean drops, median barely moves $\rightarrow$ B.

Q26. Correct Answer: C) 56

STEP-BY-STEP SOLUTION:

Total of 15: $72 \times 15 = 1,080$.

Passers (10): $80 \times 10 = 800$.

Failers (5): $1,080 - 800 = 280$.

Mean of failers: $280/5 = 56$.

WHY EACH OPTION IS WRONG/RIGHT:

A) 48 — Various calculation errors.

B) 50 — Averaged the two means: $(72+80)/2=76$... doesn't lead to 50 either.

C) 56 — CORRECT.

D) 64 — Used $72-8=64$ without weighting.

👤 **TRAP ALERT:** Averaging the two means: $(72+80)/2=76$. This ignores different group SIZES.

💡 **KEY INSIGHT:** Weighted groups: total = $\Sigma(\text{group mean} \times \text{group size})$. Never average means of different-sized groups.

⚡ **FAST METHOD:** $1080 - 800 = 280$. $280/5 = 56$.

Q27. Correct Answer: C) I and II

STEP-BY-STEP SOLUTION:

Old: 12 vals, mean=25 $\rightarrow$ sum=300, median=22.

Add 22: new sum=322, n=13, mean= $322/13 \approx 24.77$.

I: $24.77 < 25 \rightarrow$ mean decreases ✓.

II: Old median=22 (avg of 6th+7th). Adding 22 → 13 vals → median=7th. Since 22 was the old median and we're adding that exact value at the median position, new median=22 ✓.

III: Mean changed ($24.77 \neq 25$) → FALSE.

WHY EACH OPTION IS WRONG/RIGHT:

A) II is also true.

B) I is also true.

C) CORRECT.

D) III is false (mean changed).

 **TRAP ALERT:** Thinking adding a value equal to the median doesn't affect the mean. It does: $22 < 25$ pulls the mean down.

 **KEY INSIGHT:** Adding x to a set: mean changes unless x = current mean. Median stays if x = current median.

 **FAST METHOD:** $22 < 25$ → mean drops(I). $22 = \text{median}$ → stays(II). C.

Q28. Correct Answer: 82

STEP-BY-STEP SOLUTION:

Original: 20 students, mean=80 → total=1,600.

Add 2 to each: new total= $1,600 + 2(20) = 1,600 + 40 = 1,640$.

Add student scoring 82: total= $1,640 + 82 = 1,722$.

New $n=21$. Mean= $1,722/21=82$.

WHY EACH OPTION IS WRONG/RIGHT:

84 — Added 2 to the mean (82) then rounded.

81 — Forgot to add the correction to all scores.

80 — Didn't account for either change.

82 — CORRECT.

 **TRAP ALERT:** Adding 2 to the mean only (getting 82) without accounting for the new student, or adding the student without the correction.

 **KEY INSIGHT:** Handle each change step by step: (1) correction adjusts the total, (2) new student changes both total and count.

 **FAST METHOD:** $1600 + 40 + 82 = 1722$. $1722/21 = 82$.

Q29. Correct Answer: A) 20

STEP-BY-STEP SOLUTION:

Original: even count, middle values 16 and 24. Median= $(16+24)/2=20$.

Add 20: count becomes odd.

20 falls between 16 and 24 → becomes the new single middle value.

New median = 20.

WHY EACH OPTION IS WRONG/RIGHT:

A) 20 — CORRECT.

B) Can't be less — 20 is between the old middles.

C) Can't be more — 20 is between.

D) It IS always 20 — not uncertain.

 **TRAP ALERT:** Overthinking it. 20 slots between 16 and 24 and becomes the center of the new odd-count set.

 **KEY INSIGHT:** Adding a value between the two old middle values: that value becomes the new median.

 **FAST METHOD:** 20 between 16 and 24 → new median = 20.

Q30. Correct Answer: B) Mean decreases; median stays

STEP-BY-STEP SOLUTION:

Old sum = $36 \times 9 = 324$. Replace 90 with 60: new sum = $324 - 90 + 60 = 294$.

New mean = $294/9 \approx 32.67$ (↓ from 36).

Median: 5th value of 9 = 30 (unchanged — only the 9th/largest changed).

WHY EACH OPTION IS WRONG/RIGHT:

A) Median doesn't change.

B) CORRECT.

C) Mean always changes when a value changes.

D) Mean always changes.

 **TRAP ALERT:** Thinking changing the largest value affects the median. Only the MIDDLE values determine the median.

 **KEY INSIGHT:** Changes to extreme values (min or max) affect the mean but NOT the median.

 **FAST METHOD:** 90 → 60: mean drops. 5th value still 30. B.

TOPIC SUMMARY — Statistics: Mean & Median

Common Mistakes:

1. Finding median WITHOUT sorting the data first — the #1 statistics error.
2. For even-count data: forgetting to AVERAGE the two middle values.
3. Confusing mean with median, especially when outliers are present.

Must Remember:

1. Mean = $\text{sum} \div \text{count}$. Missing value = $(\text{target mean} \times n) - \text{known sum}$.
2. Median (odd n) = value at position $(n+1)/2$. Median (even n) = average of two middle values.
3. Mean is sensitive to outliers. Median is resistant. Removing an outlier changes the mean dramatically but barely affects the median.

Expected on DSAT: 2–3 questions per test. Expect one add/remove value question, one frequency table mean, and one outlier-effect question.

 **DESMOS / CALCULATOR:** Use the built-in calculator for ALL addition. Never add large sums mentally. For frequency tables: compute $\Sigma(\text{value} \times \text{freq})$ first, then divide.